

The Twenty-Eight Mansions on the Lacquer Box from the Tomb of Marquis Yi of Zeng: A Lower Bound for the Four Images

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2026-10-03 · 预印本 (preprint) · 未经同行评议 · v2

DOI: 10.5281/zenodo.23032029

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Abstract

In 1978, at Leigudun in Suixian, Hubei, present-day Suizhou, an archaeological team lifted a wooden box out of a Warring States tomb. The lid was taken off: on a ground of black lacquer a ring of characters had been painted in vermilion, twenty-eight of them, with a large seal-script 「斗」, the Dipper, in the middle. Outside the ring, one at each end, a dragon was painted on the east and a tiger on the west. The box is now known as the twenty-eight-mansion diagram clothes box. No account of where the twenty-eight mansions came from can get around it, because it is early; and one further sentence is habitually added to it: on this box there are the twenty-eight mansions, and there are also the four images.

That sentence is worth weighing. What an excavated object supplies is a *terminus ante quem*, a lower bound, at the latest; not a *terminus post quem*, from when. The twenty-eight names on this lid can still be read one by one, and the dragon and the tiger are plainly drawn. But "four images" means four creatures, and on this lid only two are visible. Beneath the term lie three separate layers: a closed list of twenty-eight names, the division of that list into four groups of seven, and the pairing of each group with one animal. The three need not arrive together. Working backwards from the box, the earliest hard evidence for animal-plus-direction is the Xishuipo tomb at Puyang, Henan, whose clam-shell dragon, tiger and Dipper are dated by radiocarbon measurement on shell, tree-ring corrected, to 6460 ± 135 years before present, and which contains not one star name. The four creatures are written down side by side only in the Eastern Han. The lower bounds therefore lie at three different depths, from about six and a half

thousand years ago down to the Eastern Han, and the lacquer box reaches only the second of them. The piece is descriptive and carries no prediction.

Keywords: Marquis Yi of Zeng; lacquer box; twenty-eight mansions; four images (四象); Xishuipo; terminus ante quem; Warring States astronomy; Han astronomy

Introduction

In 1978, at Leigudun in Suixian, Hubei, present-day Suizhou, an archaeological team lifted a wooden box out of a Warring States tomb. The lid was taken off: on a ground of black lacquer a ring of characters had been painted in vermilion, twenty-eight of them, and in the middle a large seal-script 「斗」. Outside the ring, one at each end, a dragon was painted on the east and a tiger on the west.

The box came to be known as the twenty-eight-mansion diagram clothes box. No account of where the twenty-eight mansions came from can get around it, because it is early.

Getting around it is not the only thing that happens to it. One further sentence is habitually added: on this box there are the twenty-eight mansions, and there are also the four images.

That sentence is worth pausing over. The twenty-eight characters on the box can still be read one by one, and the dragon and the tiger are drawn plainly. But "four images" is four things, dragon, tiger, bird, turtle-and-snake, and on the lid only two are visible. The other two do not appear.

There is a further layer. The date of the box is generally put at about 433 BCE. From the sentence above, some go on to infer: the Chinese system of four images was already complete in the early Warring States.

That is a step taken a little quickly. What follows takes the sentence apart and asks how much weight it can carry.

1. What one object can pin down

In archaeological reporting there is one sentence that is often passed over: what a single object gives you is "at the latest".

Suppose a wooden tablet is dug out of the ground with 「正月」 written on it, and it measures to two thousand three hundred years old. What can be said is: at least two thousand three hundred years ago, someone was already calling a month by this name. What cannot be said is: this way of naming began in that year. By the time the tablet was put down, the usage had most likely been going for some while. How long, the tablet does not answer.

In chronology the first of these is called the lower bound, or the latest possible date; the second is called the upper bound, or the earliest possible. A single object can usually pin down only the lower bound, never the upper.

The lower bound by itself is already valuable. It compresses the year of invention of a system from "unknown" to "no later than such-and-such". The trouble begins at the moment the lower bound is used as an upper bound, when "at the latest there was already" is read as "only from this year was there".

This is the third piece in the series. The first two asked whether something could be computed, and which few records ought never to be computed; this one changes direction. It asks which end of a system a physical object can pin down, and how the stretch it cannot pin down ought to be described.

2. The box

The look of the box is not described in detail here. What matters is a few details, because they decide how much the opening sentence can bear.

First, an outline. This is a wooden clothes box, lid and body each hollowed out of a single block of wood; the lid is arched, the interior red-lacquered, the exterior black-lacquered as ground with vermilion patterns. Length 71 cm, width 47 cm, overall height 40.5 cm. Five clothes boxes of similar type came out of the same tomb. The one carrying the star diagram is numbered E.66; among the others, E.61 is painted with Hou Yi shooting the suns and E.67 with Kuafu chasing the sun.

On the face of the lid, at the exact centre, is a seal-script 「斗」. Around this character runs a ring of twenty-eight characters, written in the local Chu variant forms, from 角, 亢, 氐, 房, 心, 尾, 箕 all the way round to 軫. Outside the ring, on the east and west sides, a dragon is painted on one side and a tiger on the other. The two end faces of the lid each carry a group of ornament; of the two long sides, one is painted with two beasts facing off, and the other carries only a red band.

There are several places where a reader does well to stop.

The first is the character 「斗」. Some take its strokes apart by direction: the two ends of the horizontal point to 心 and 危 respectively, and the end of the vertical points to 張, while the handle end points to 觜觿. If one follows the actual sky, the place the handle points to ought rather to fall at 畢. It points wrongly. This is often used to show that the diagram on the lid is not necessarily traced from the real sky of one particular night, and is more like a diagram drawn as a diagram.

The second is four small characters beside 亢: 「甲寅三日」. Some compare them with the old calendars and hold that this figure depicts the sky of the evening of the third day of the first month in 433 BCE, days that fell around the beginning of spring, when the ancient rites required the ploughing ceremony 【一说】. Which year and which event these four characters refer to has other readings as well.

The third is the direction of the sequence. The twenty-eight characters on the lid run clockwise, the reverse of later star charts. Guo Dewei offered an explanation: the painter took the arched lid as the round vault of heaven and the rectangular base of the box as the earth; a person standing on the earth looks up at the vault, and looking at this lid from above reverses the order. On this account the difference in direction is really the difference between looking down and looking up.

Put the three together and the identity of the object is fairly clear. It is an inscription on a vessel, not an observation log.

The date of the box itself has two values. 433 BCE is the one in general use 【通说】, and there is a further proposal of 435 BCE 【一说】. The two numbers differ by two years, which makes no difference to the judgement this piece has to make; the one in general use is taken here, and the matter is not put too firmly.

What it can pin down is this one sentence: at the latest, around 433 BCE, the twenty-eight star names together with their order, plus one 「斗」, plus a dragon to the east and a tiger to the west, were already common knowledge, painted casually on the lid of a box for clothes.

As to what it cannot pin down, the term "four images" has first to be taken apart.

3. Beneath the four images lie three layers

"Four images" on the side of the sky means the division of the twenty-eight mansions into four groups by the four directions, each paired with one animal: azure dragon in the east, white tiger in the west, vermilion bird in the south, dark warrior in the north.

This can in fact be split into three layers.

The first: twenty-eight names gathered into one closed list.

The second: the list cut by the four directions into four groups of seven.

The third: each of the four groups paired with one animal.

The three need not arrive together. They may appear one after another, or stop at the second. When the opening piece of this series discussed the four fixed stars of the 「Yaodian」, it noted

that those four stars each stand within one image; that is a configuration which exists only after all three layers are in place.

Take the first layer first. The gathering of twenty-eight names into one closed list happened very late. Before that, star names in the literature are scattered: in the Yinxu divination inscriptions there is the Bird star and there is the Great Fire, stars the Shang people already knew and used for keeping records; in the *Shijing* and the *Xia Xiaozheng* there is 箕, there is 斗, there is 辰, there is 參, scattered here and there. They are all there; they simply do not form a file. The previous piece, saying that the men of Shang knew a good many stars but had no twenty-eight-mansion coordinates, was talking about exactly this stretch. The grid was drawn in later; how much later, a lower bound can be seen below.

Now search towards the earliest end.

In 1987, at Xishuipo in Puyang, Henan, in a tomb of the Yangshao culture, a dragon was laid out in clam shell to the east of the occupant, a tiger to the west, and between them a Dipper made of clam shell and human bone.

The date of this tomb requires a set of numbers here. The laboratory of the Institute of Archaeology, Chinese Academy of Social Sciences, made radiocarbon measurements on clam shell from the tomb; corrected by tree rings, the result is 6460 ± 135 years. A second set, inferred from the Hougang type, falls between 4300 and 4100 BCE. Taking the radiocarbon set, the text writes "about six thousand five hundred years ago", with the error recorded as ± 135 years.

It has no star names, not one. What Xishuipo can say is this: the dragon and the tiger, these two images, were laid out to east and west very early, with the Dipper between them. That is the hardest single piece of evidence for directions being laid out; what it governs has nothing to do with the twenty-eight mansions.

From this tomb to Marquis Yi of Zeng is a gap of about four thousand years. Four thousand years ago there were dragon, tiger and Dipper; four thousand years later there are twenty-eight names plus dragon, tiger and Dipper. Join the two points and a very long line can be drawn: the pairing of animals with directions is very old, and was never broken.

The middle of the line is empty.

4. Afterwards: in how many steps do the four animals arrive

On the lid of Marquis Yi's box, two can be seen at a glance. The completion of four beasts has to be sought in later material. That stretch of road is long.

Look first at the numbers in the classics. In the 「Zhouli · Chunguan」 there is an office called Fengxiang shi, whose charge is 「十有二歲，十有二月，十有二辰，十日，二十有八星之位」. This sentence was quoted in the preceding instalment; it is borrowed here to say something else. The classics have the number "twenty-eight", and never once list the twenty-eight names. The number arrived; the list did not.

Look next at the list. In the transmitted literature the twenty-eight names appear together in a batch for the first time in the Lüshi Chunqiu · Youshi, at the end of the Warring States:

「天有九野……中央曰鈞天，其星角、亢、氐；東方曰蒼天，其星房、心、尾……」

The names are complete. But look at the division: nine parts. One central, four for the four directions, four for the four corners, together cutting the twenty-eight mansions into nine shares. The cut of seven to each of the four directions does not yet exist in this text.

One further thing is worth noticing. In the nine-field division the central share receives 角, 亢, 氐, and the eastern share receives 房, 心, 尾. Three mansions to a share, four to a share, which does not agree with the neat later cut of seven to each direction. The nine fields govern directions and field-allotments; what they divide is not the four images.

In the same stretch of years there are other versions. In the eleventh volume of the Tsinghua bamboo slips, published in 2021, there is a text called 「五紀」, a Chu document of the middle to late Warring States, and it too contains a full set of twenty-eight mansions. According to the editors' transcription it is not entirely the same as the set on the lacquer box: in the western seven it writes 「濁」 rather than 「畢」; 參 and 伐 are two separate mansions and are not merged into one; in the south it uses 狼 and 弧 rather than 井 and 鬼.

By the Han the matter becomes more tangled rather than less.

The Huainanzi · Tianwenxun, in the early Western Han, sets out five directions, each with one beast:

「東方，木也……其獸蒼龍」「南方，火也……其獸朱鳥」「中央，土也……其獸黃龍」「西方，金也……其獸白虎」「北方，水也……其獸玄武」

Count them: five beasts. The centre has an extra yellow dragon.

Further on, in the Shiji · Tianguanshu, Sima Qian writes the four palaces, and begins with four sentences:

「東宮蒼龍，房、心。」「南宮朱鳥，權、衡。」「西宮咸池，曰天五潢。」「北宮玄武，虛、危。」

The eastern palace is the azure dragon, the southern palace the vermilion bird, the northern palace the dark warrior; the western palace, however, is written 「咸池」. Xianchi is the name of an asterism, not a beast. The two characters 「白虎」 occur elsewhere in the Tianguanshu, attached beneath 參; they are the image of that mansion, not the name of the western palace.

That is to say: down to the early Western Han, the western palace had still not been exchanged in writing for the name of a beast.

The four written side by side are fixed only by the Eastern Han. In Zhang Heng's Lingxian there is a sentence:

「蒼龍連蜺於左，白虎猛據於右，朱雀奮翼於前，靈龜圈首於後。」

At this point four directions, four animals and one fixed arrangement have finally reached the written record.

5. So what is the lower bound

Set the material above into three layers and the answer comes out by itself.

Layer	Latest date	Basis
Dragon and tiger to east and west, Dipper between them	6460±135 years before present (clam-shell radiocarbon, tree-ring corrected)	The shell figures of Xishuipo tomb 45, Puyang
The twenty-eight names and their order as one set	about 433 BCE	Lacquer box E.66, tomb of Marquis Yi of Zeng
Twenty-eight mansions divided by the four directions into four groups, each with one beast	Eastern Han	Zhang Heng's Lingxian; in the Western Han Shiji the western palace is still 「咸池」

Three dates, spanning from six and a half thousand years ago to the Eastern Han. They do not conflict, because they are talking about three different things.

Go back to the opening sentence: this box has the twenty-eight mansions, and it has the four images too. Strictly speaking, what it can support is the first half. The box gives the twenty-eight names, gives the Dipper, gives the dragon and the tiger; if "four images" is to mean four beasts complete, that must wait several centuries longer.

One reading does not accept this limitation. Huang Jianzhong and others opened the box's six faces out flat and held that the pattern on the south face is precisely the bird representing the south, and that the north face, blackened over, represents the dark warrior. Were that reading

to hold, the four images would in fact be complete on the lacquer box, only distributed over different faces instead of on the lid.

This is 【一说】, and it rests on the identification of patterns. That the south face shows a bird can be argued; whether a black face counts as the dark warrior depends even more on inference. Direct evidence in writing is not this roundabout; a pattern has to be read by eye and argued. The two kinds of evidence do not weigh the same, and in reading they must be counted separately. Where the evidence is a pattern, the reading drawn from it is 【一说】.

So the more accurate formulation is this: the lacquer box from the tomb of Marquis Yi of Zeng pins the lower bound of the twenty-eight-mansion system to the early Warring States, and at the same time supplies the dragon and the tiger, so that the line of "beast paired with direction" can be joined up from below. The step at which the four beasts are complete must wait, in objects and in texts alike.

Henceforth, whenever one meets "so-and-so proves that China had the four images as early as such-and-such a year", three questions can be put first: which layer does it mean; does it rest on writing or on pattern; and is what it gives a latest date, or a beginning.

The twenty-eight names on the lid were put there for people to look up and identify the sky. But to fix a date by them, one thing is still missing, and that is measuring the length of the shadow each day.

Note on the text

This is a popular exposition, and it makes no claim to original research. The astronomical constants cited are standard values. This English version is issued alongside the Chinese original as the same article in another language. The author's name is 邝楚嘉, Chujia Kuang; he writes under the pen-name 嘉言一得 (Jiayan Yide).

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record.

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